

TrES-3b

R-2730

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160624 · created 2026-09-02T00:20:28+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457563.7091 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.85 +/- 2.24 [%]
Orbital Inclination (inc)	86.84 +/- 2.87
Airmass coefficient 1 (a1)	17.66 +/- 12.55
Airmass coefficient 2 (a2)	-2.83 +/- 0.97
Scatter in the residuals of the lightcurve fit is	65.43 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	4.5
Transit Duration (day)	0.08 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.057 vs 0.16309 (deviation 0.58σ)
Duration fit vs published	0.08 d vs 0.0567 d (deviation 2.12σ)
Mid-time O–C	-0.4 min
Depth SNR	3.4
Residual scatter	65.43 %

Light curve

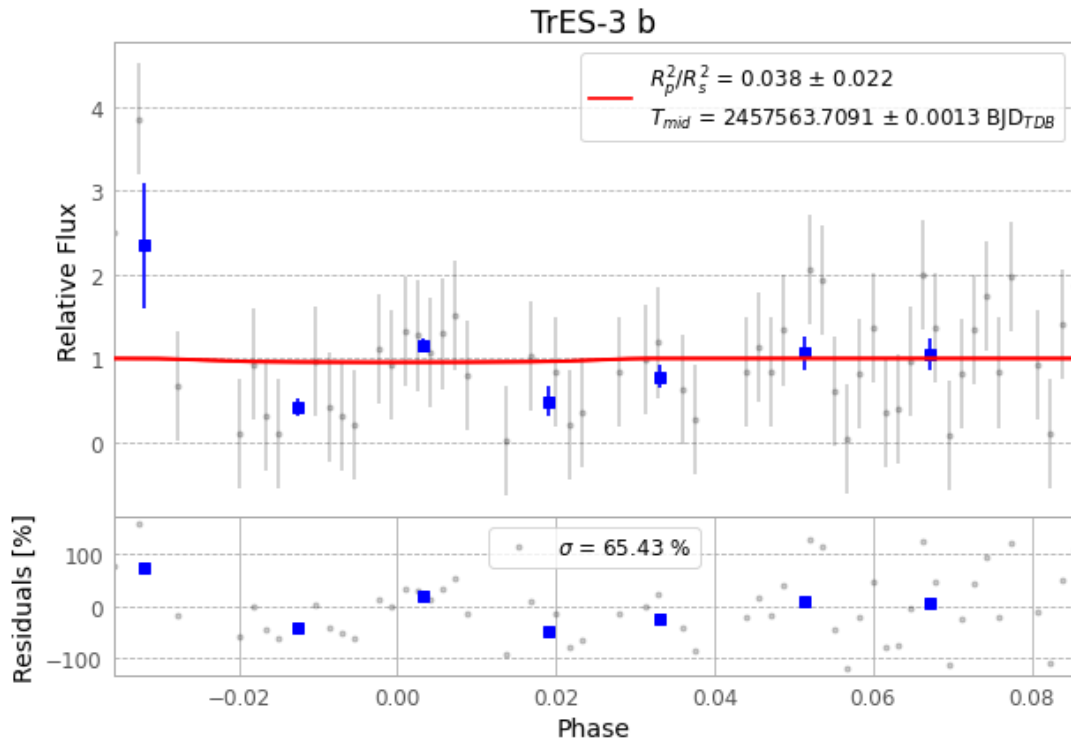


Figure 1 — FinalLightCurve_TrES-3 b_2016-06-23.png (SHA-256 1524d83e5616ba0d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 182], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 c58eaf31582716f2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

78 FITS frames (51 MB), TRES-3160624034812.FITS → TRES-3160624073912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160624.log (ea4327bf3d0e...), FinalLightCurve_TrES-3 b_2016-06-23.png (1524d83e5616...), FinalLightCurve_TrES-3 b_2016-06-23.csv (3db8e545aca7...)

Compute resources

Wall time 262.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:20Z.